Matrix Transformation

Matrix of a Linear Transformations

Properties of the Matrix-Vector Product ($A\mathbf{x}$)

If A is an $m \times n$ matrix, $\mathbf{u}$ and $\mathbf{v}$ are vectors in $\mathbb{R}^n$, and c is a scalar, then:

a. $A(\mathbf{u} + \mathbf{v}) = A\mathbf{u} + A\mathbf{v}$;

b. $A(c\mathbf{u}) = c(A\mathbf{u})$.

Linear Transformation (Defined from the above properties)

A transformation (or mapping) T is **linear** if:

(i) $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ for all $\mathbf{u}, \mathbf{v}$ in the domain of T;

(ii) $T(c\mathbf{u}) = cT(\mathbf{u})$ for all scalars c and all $\mathbf{u}$ in the domain of T.

Note: Linear transformations preserve the operations of vector addition and scalar multiplication.

Matrix of a Linear Transformations

Linear Transformations

- Reflections
- Projections
- Deformations
  - Dilations & Contractions
  - Shearing
- Rotations

Note: **Linear transformation** focuses on property of mapping.

Matrix Transformation describes how such a transformation is implemented.

Matrix of a Linear Transformations

Reflections

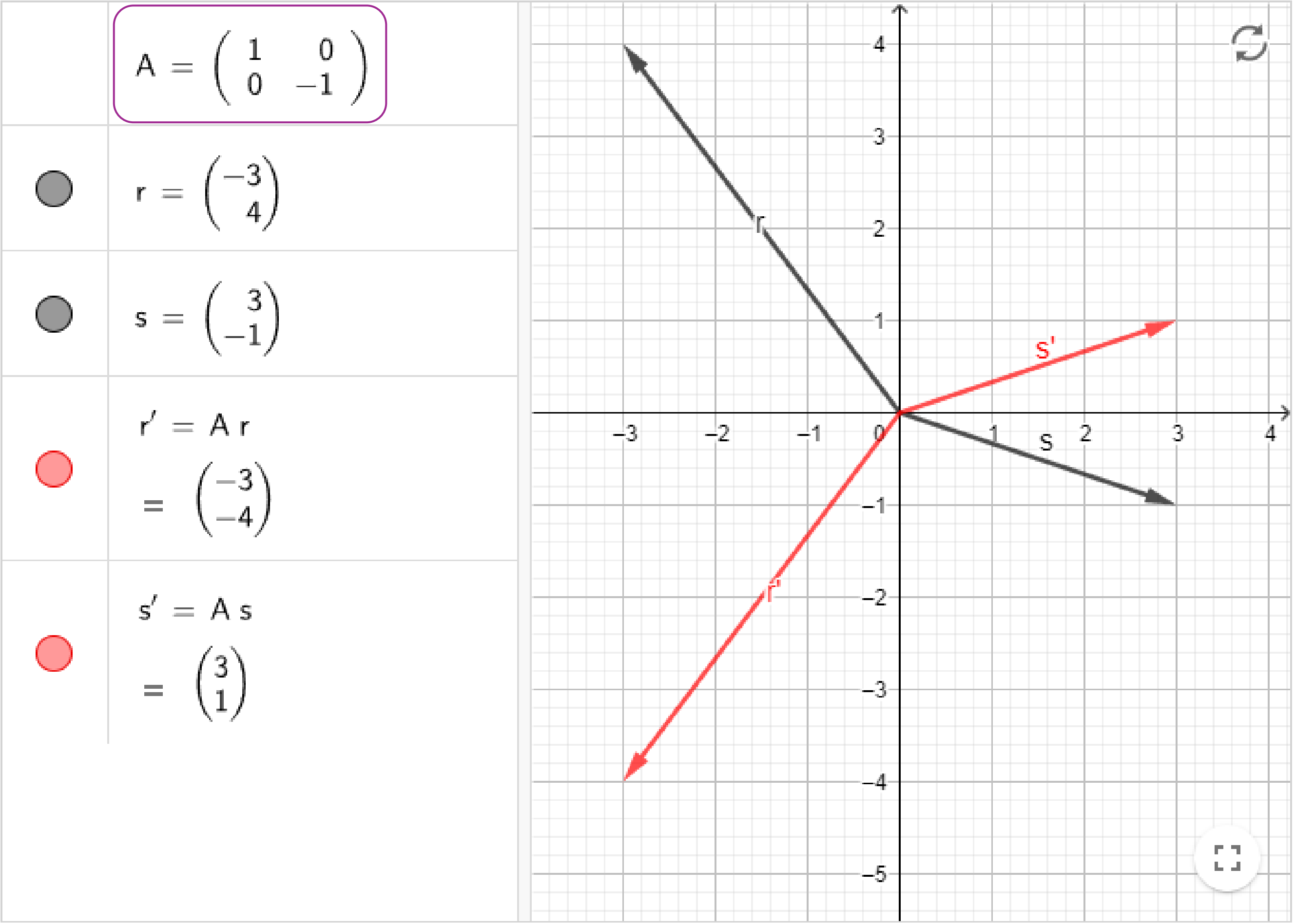
Check the transformations by following 3 transformation matrices

$$\boxed{A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}}$$ X axis reflection

$$A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$ Y axis reflection

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$ x=y line reflectino

Go to GeoGebra.org to try yourselves ⟹



Matrix of a Linear Transformations

Reflections (Continued...)

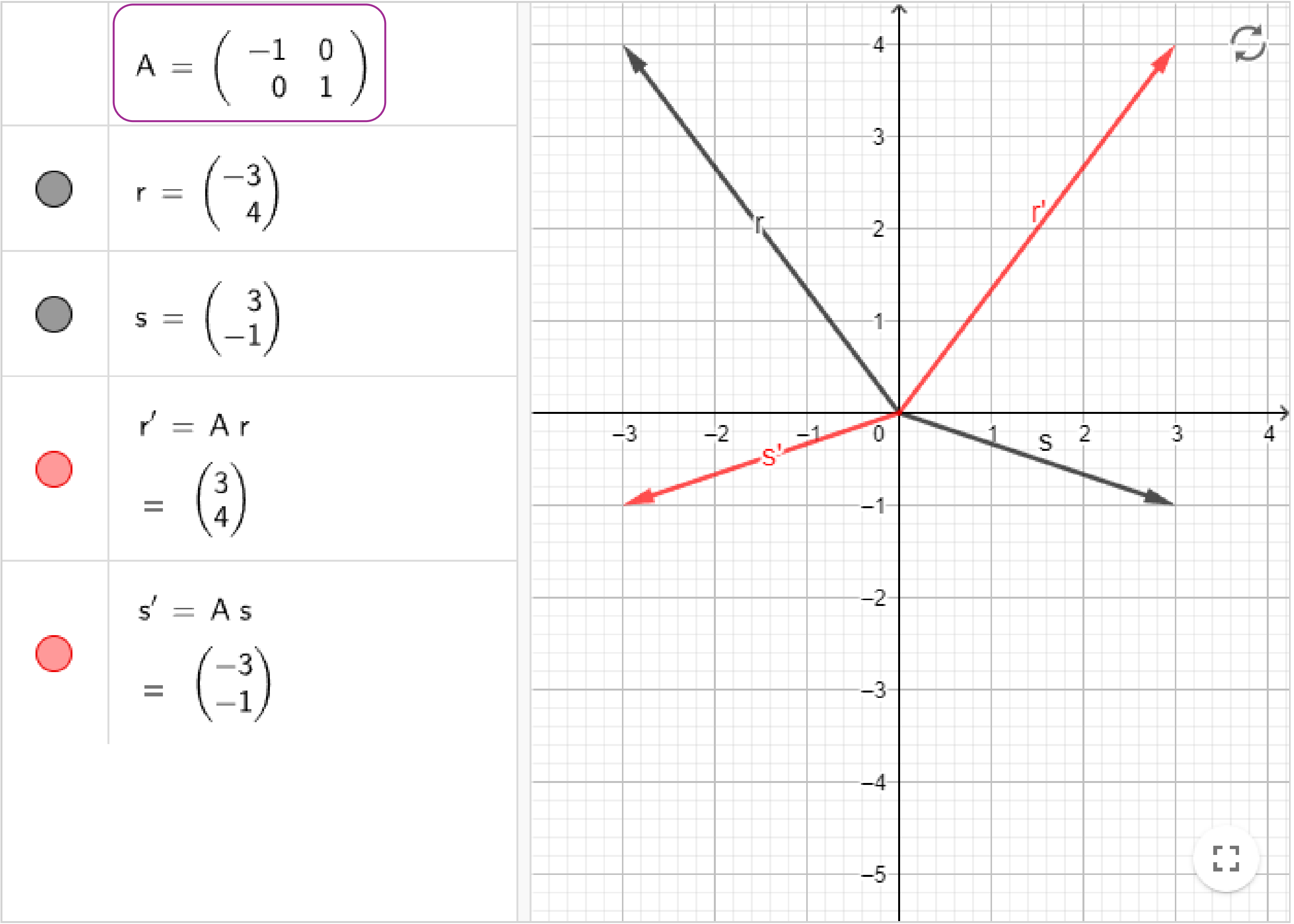
Check the transformations by following 3 transformation matrices

$$A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\boxed{A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}}$$

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Go to GeoGebra.org to try yourselves $\Longrightarrow$



Matrix of a Linear Transformations

Reflections (Continued...)

Check the transformations by following 3 transformation matrices

$$A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\boxed{A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}}$$

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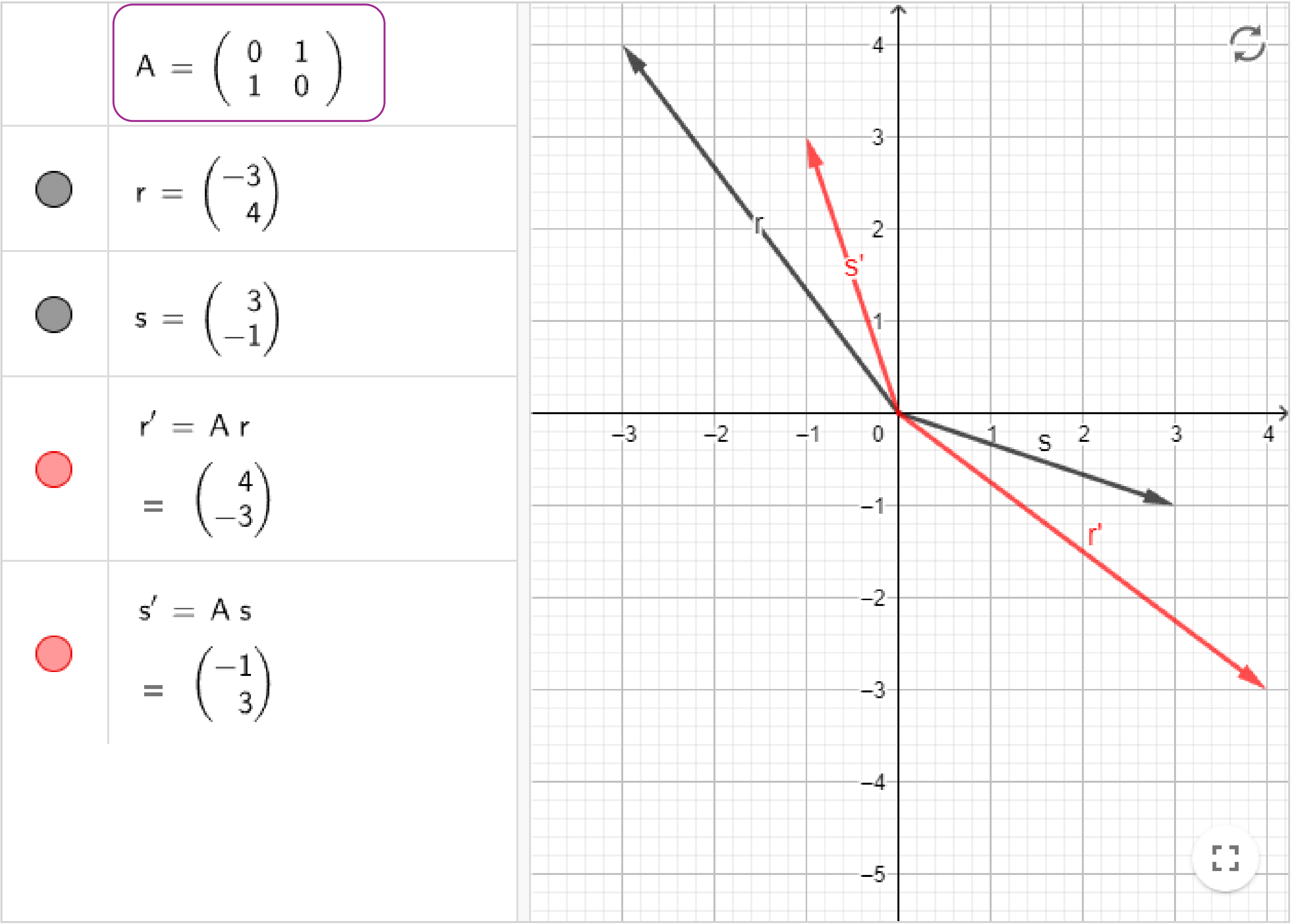


Matrix of a Linear Transformations

Projection Transformation (Orthogonal Projection)

The transformation $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^2$ defined by

$T(\mathbf{x}) = A\mathbf{x}$ is called a **projection transformation**.

Example: the function which maps the point (x,y,z) in three-dimensional space $\mathbb{R}^3$ to the point $(x,y,0)$ is an orthogonal projection onto the xy-plane.

This function is represented by the matrix:
$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

The action of this matrix on an arbitrary vector is:
$$P\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ y \\ 0 \end{bmatrix}$$

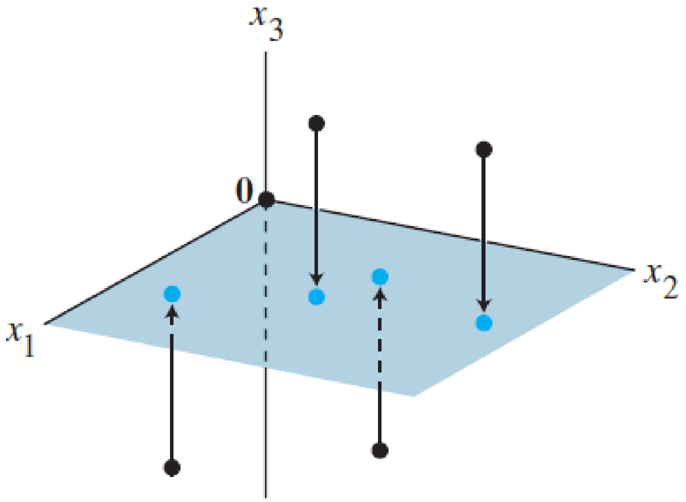
Projection transformations appear in computer graphics.

Matrix of a Linear Transformations

EXAMPLE 2 If $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$, then the transformation $\mathbf{x} \mapsto A\mathbf{x}$ *projects* points in $\mathbb{R}^3$ onto the x_1x_2-plane because

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \mapsto \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ 0 \end{bmatrix}$$

This is called "a projection transformation"



Matrix of a Linear Transformations

Shear Transformation

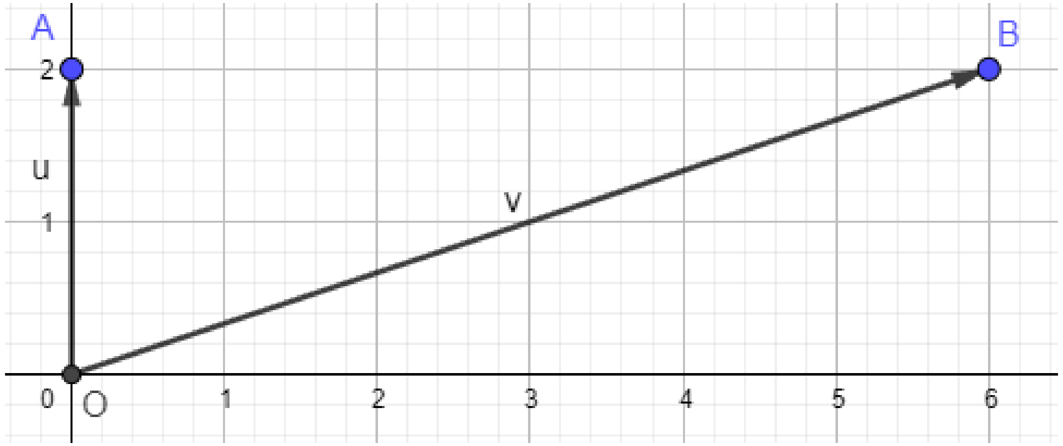
The transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ defined by $T(\mathbf{x}) = A\mathbf{x}$ is called a **shear transformation.**

dialation(expand)
+ contract
= shearing

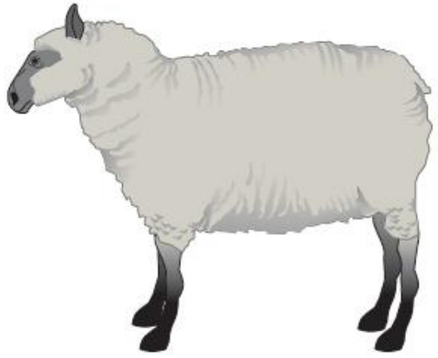
shearing factor

Example: Given

$$\mathbf{u} = \begin{bmatrix} 0 \\ 2 \end{bmatrix} \quad T(\mathbf{u}) = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$$



1 k
0



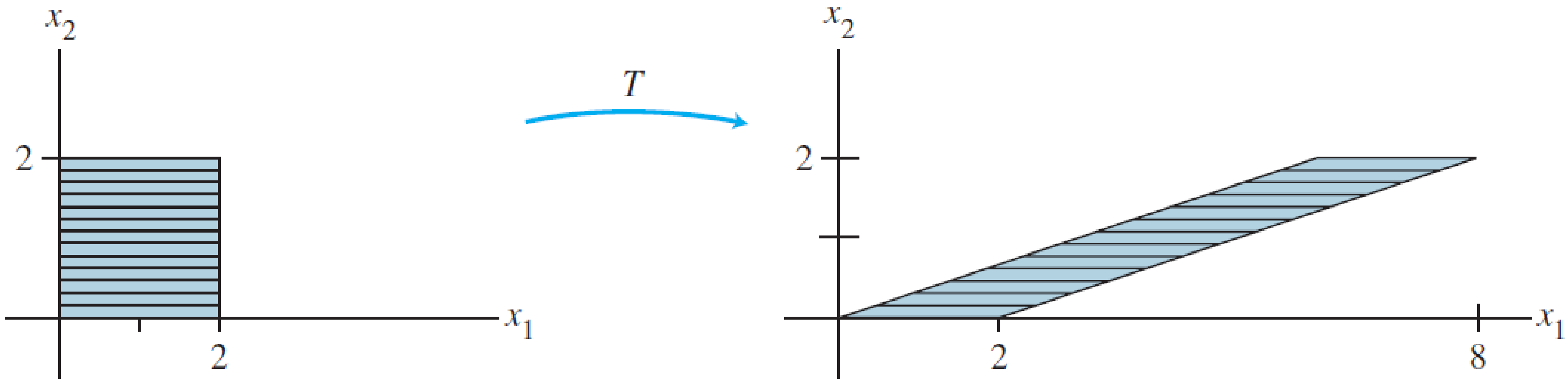
sheep

sheared sheep

Shear transformations appear in physics, geology, and crystallography.

Matrix of a Linear Transformations

EXAMPLE 3 Let $A = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$.



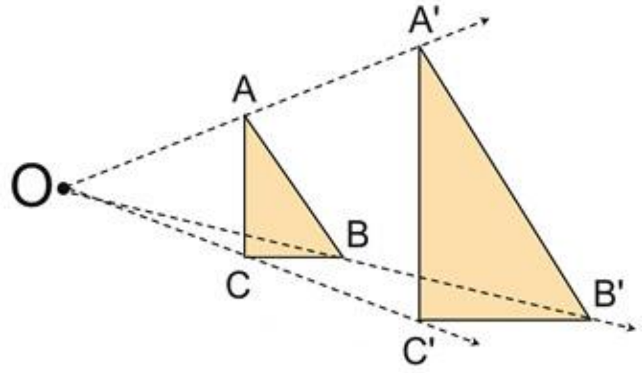
If T acts on each point in 2×2 square as shown above, then the set of images forms the shaded parallelogram (Note that, the T maps line segments onto segments).

Matrix of a Linear Transformations

Dilation Transformation

A dilation is a transformation that produces an image that is the same shape as the original but is a different size.

- A dilation that creates a larger image is called an **enlargement**.
- A dilation that creates a smaller image is called a **reduction or contraction**.
- A dilation stretches or shrinks the original figure.

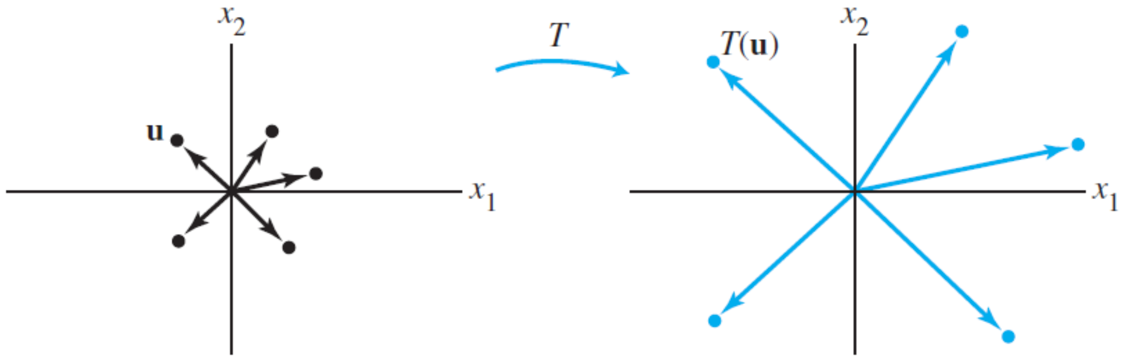


Matrix of a Linear Transformations

EXAMPLE Given a scalar r, define $T : \mathbb{R}^2 \to \mathbb{R}^2$ by $T(\mathbf{x}) = r\mathbf{x}$. T is called a **contraction** when $0 \le r \le 1$ and a **dilation** when $r > 1$. Let $r = 3$, and show that T is a linear transformation.

SOLUTION Let $\mathbf{u}$, $\mathbf{v}$ be in $\mathbb{R}^2$ and let c, d be scalars. Then

$$\begin{aligned} T(c\mathbf{u} + d\mathbf{v}) &= 3(c\mathbf{u} + d\mathbf{v}) && \text{Definition of } T \\ &= 3c\mathbf{u} + 3d\mathbf{v} && \rbrace\ \text{Vector arithmetic} \\ &= c(3\mathbf{u}) + d(3\mathbf{v}) && \\ &= cT(\mathbf{u}) + dT(\mathbf{v}) \end{aligned}$$



Matrix of a Linear Transformations

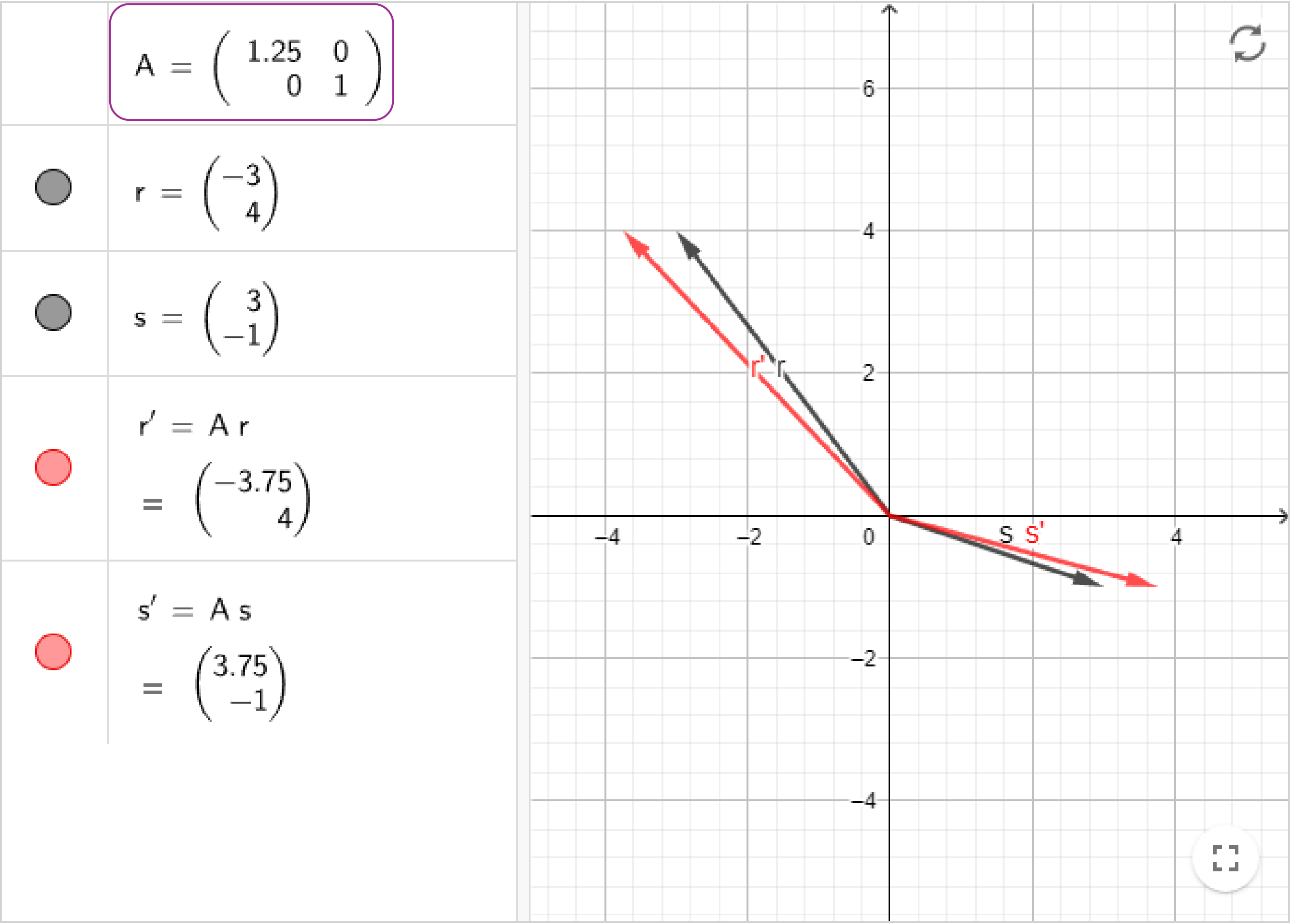
Dilations

Check the dilations by following

2 transformation matrices

$$\boxed{A = \begin{bmatrix} k & 0 \\ 0 & 1 \end{bmatrix}}$$ $\vec{v}'$ is a horizontal expansion when k>1

$$A = \begin{bmatrix} 1 & 0 \\ 0 & k \end{bmatrix}$$ $\vec{v}'$ is a vertical expansion when k>1



Matrix of a Linear Transformations

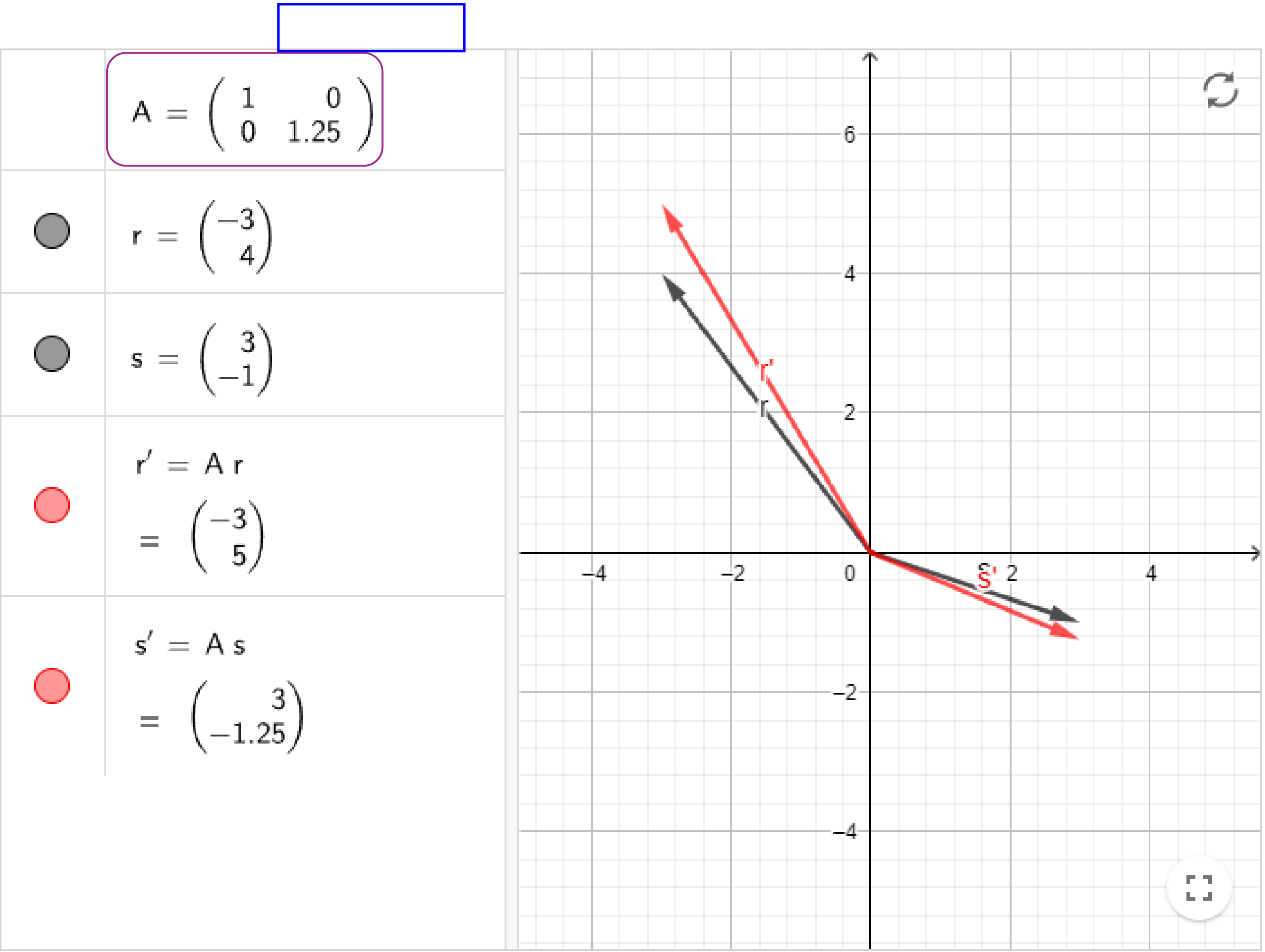
Dilations (Continued...)

Check the dilations by following

2 transformation matrices

$A = \begin{bmatrix} k & 0 \\ 0 & 1 \end{bmatrix}$ $\vec{v}'$ is a horizontal expansion when k>1

$\boxed{A = \begin{bmatrix} 1 & 0 \\ 0 & k \end{bmatrix}}$ $\vec{v}'$ is a vertical expansion when k>1



Matrix of a Linear Transformations

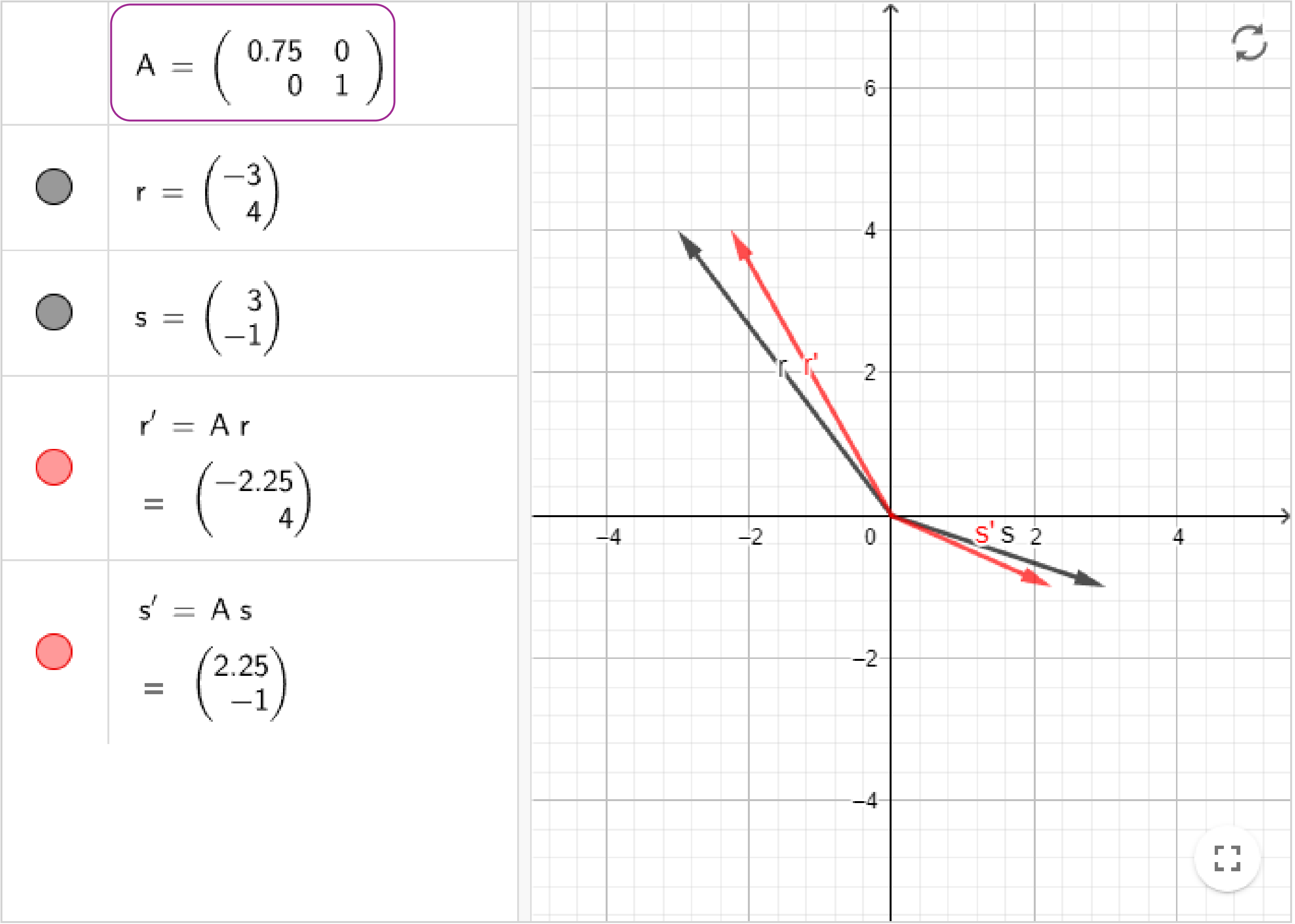
Contractions

Check the contractions by following

2 transformation matrices

$$\boxed{A = \begin{bmatrix} k & 0 \\ 0 & 1 \end{bmatrix}}$$ $\vec{v}'$ is a horizontal contraction when 0<k<1

$$A = \begin{bmatrix} 1 & 0 \\ 0 & k \end{bmatrix}$$ $\vec{v}'$ is a vertical contraction when 0<k<1



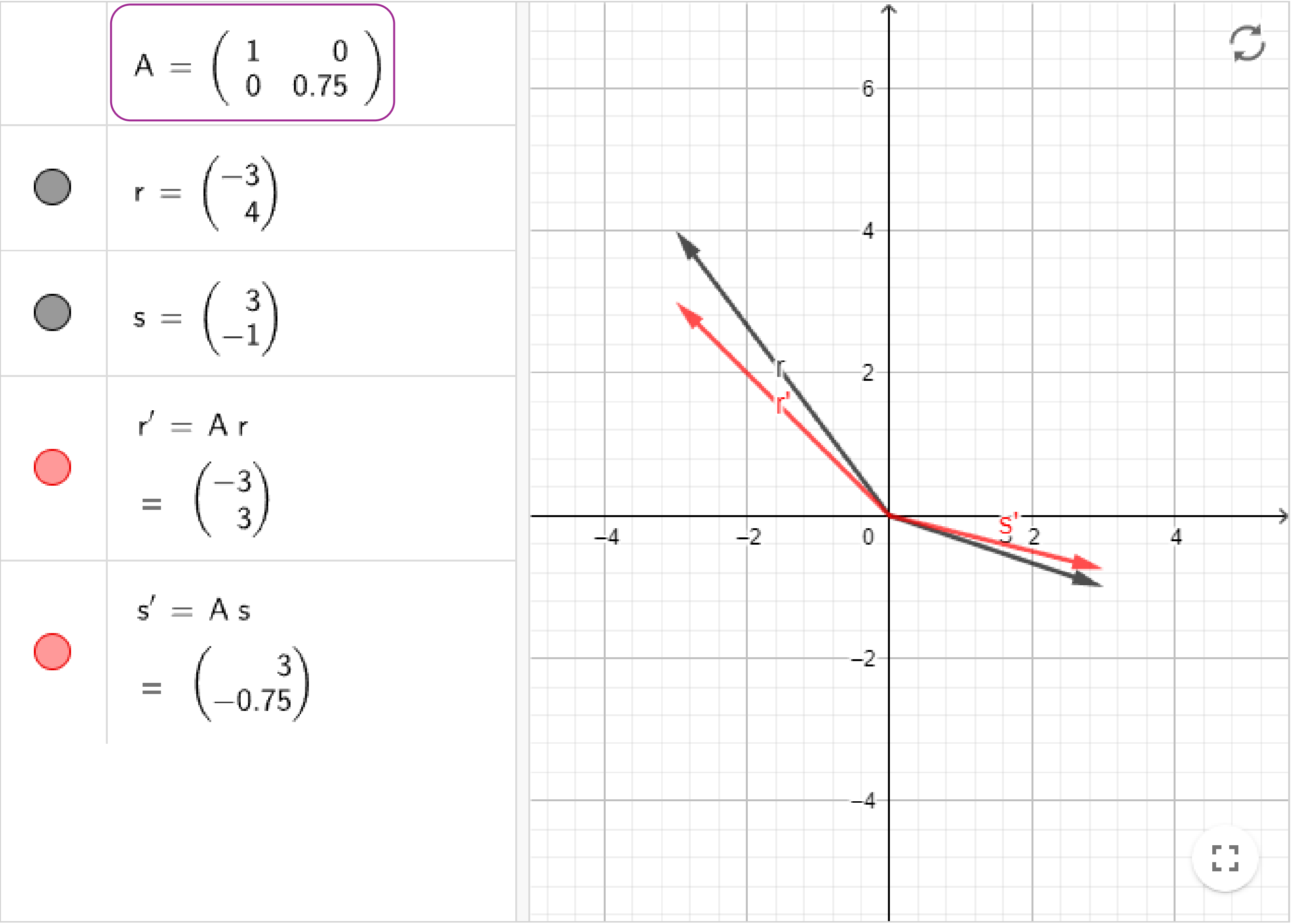
Matrix of a Linear Transformations

Contractions (Continued...)

Check the contractions by following 2 transformation matrices

$A = \begin{bmatrix} k & 0 \\ 0 & 1 \end{bmatrix}$ $\vec{v}'$ is a horizontal contraction when $0<k<1$

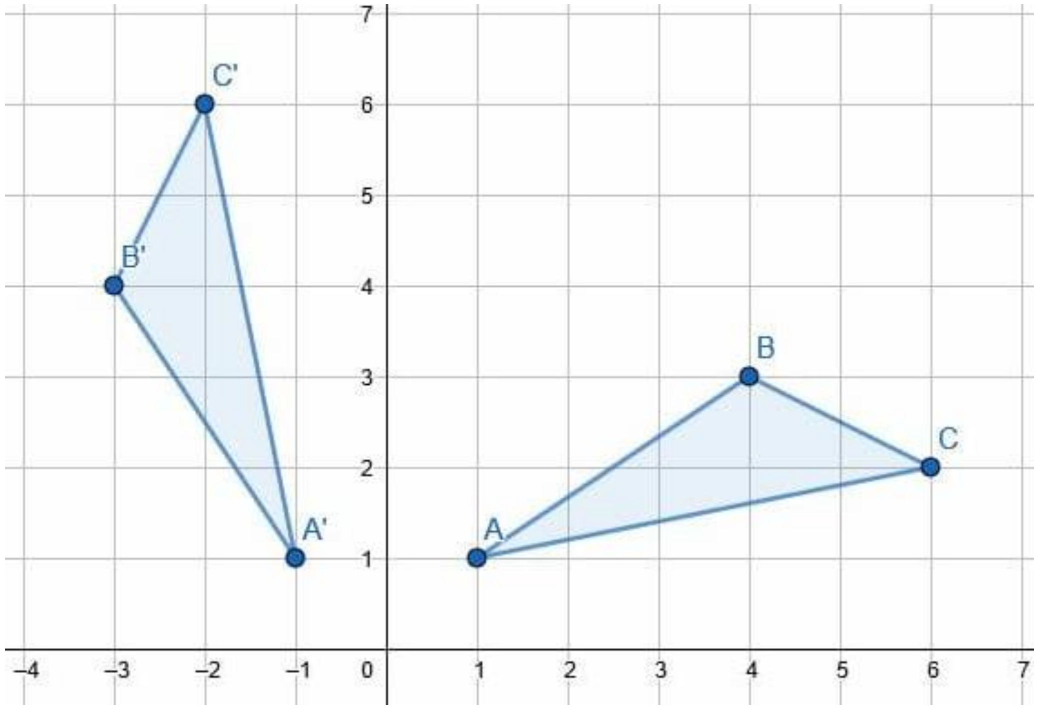
$\boxed{A = \begin{bmatrix} 1 & 0 \\ 0 & k \end{bmatrix}}$ $\vec{v}'$ is a vertical contraction when $0<k<1$



Matrix of a Linear Transformations

Rotation Transformation

A rotation is a transformation in which the object is rotated about a fixed point.

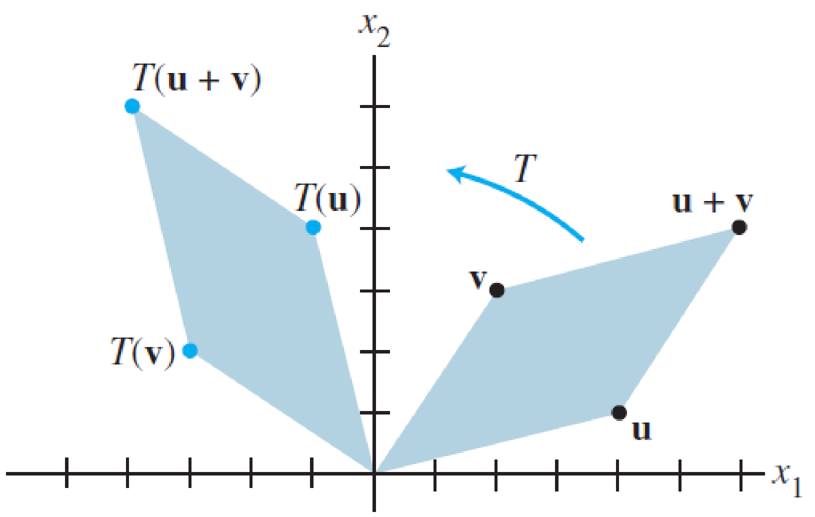


Matrix of a Linear Transformations

EXAMPLE Define a linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ by

$$T(\mathbf{x}) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -x_2 \\ x_1 \end{bmatrix}$$

Find the images under T of $\mathbf{u} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$, and $\mathbf{u} + \mathbf{v} = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$.



SOLUTION

$$T(\mathbf{u}) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 4 \end{bmatrix}, \qquad T(\mathbf{v}) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} -3 \\ 2 \end{bmatrix},$$

$$T(\mathbf{u} + \mathbf{v}) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 6 \\ 4 \end{bmatrix} = \begin{bmatrix} -4 \\ 6 \end{bmatrix}$$

Note that $T(\mathbf{u} + \mathbf{v})$ is obviously equal to $T(\mathbf{u}) + T(\mathbf{v})$. It appears from Fig. that T rotates $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{u} + \mathbf{v}$ counterclockwise about the origin through 90°.

Matrix of a Linear Transformations

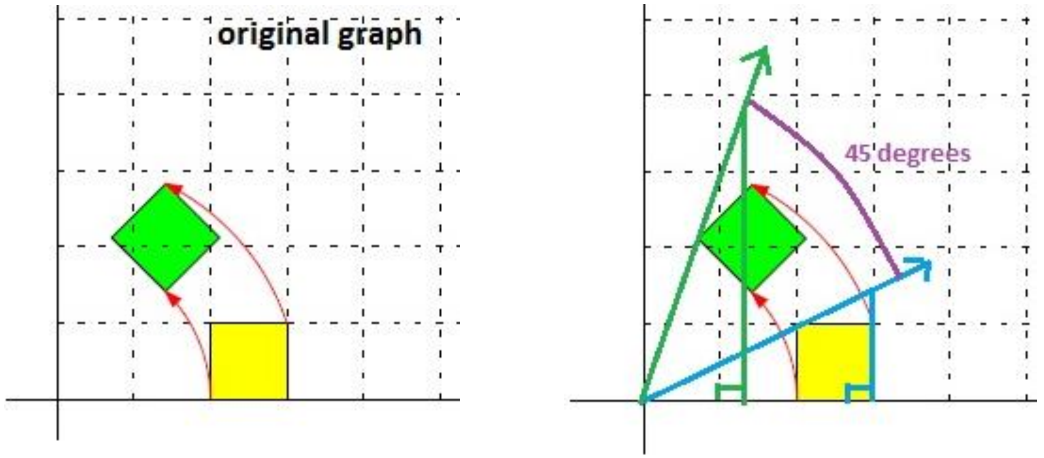
Rotation Transformation (Continued...)

To rotate 45° about the origin, we apply the matrix

$$\begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix} = \frac{\sqrt{2}}{2}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$$

Note: $\frac{\sqrt{2}}{2} = \cos 45^\circ = \sin 45^\circ$, so this is the same as

$$\begin{pmatrix} \cos 45^\circ & -\sin 45^\circ \\ \sin 45^\circ & \cos 45^\circ \end{pmatrix}$$



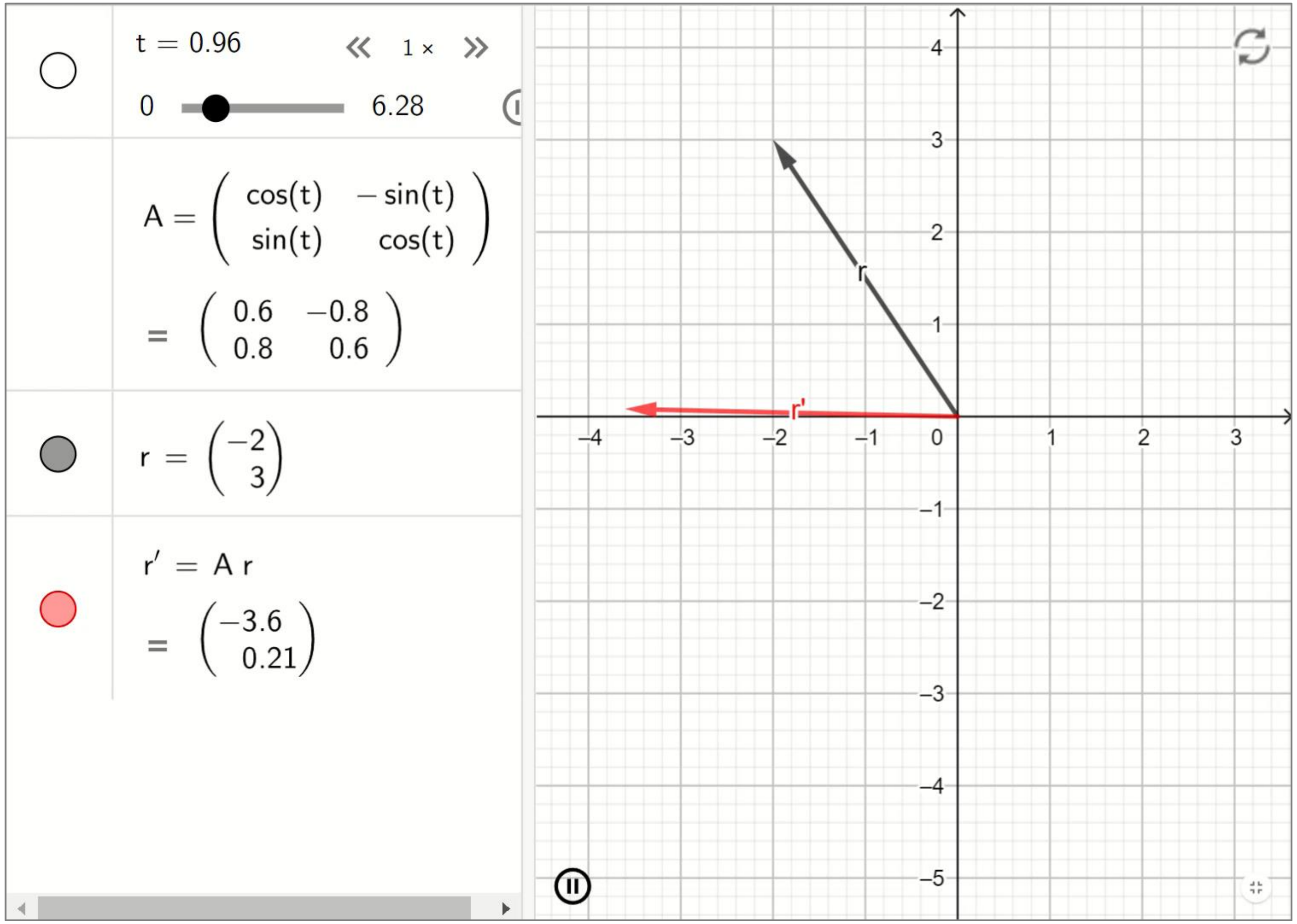
Clockwise $\begin{pmatrix} \cos\phi & \sin\phi \\ -\sin\phi & \cos\phi \end{pmatrix}$

Counter Clockwise $\begin{pmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{pmatrix}$

Matrix of a Linear Transformations

Rotation Transformation

(CCW Rotation)



Use the matrix below for a CW rotation

$$\begin{pmatrix} \cos(t) & \sin(t) \\ -\sin(t) & \cos(t) \end{pmatrix}$$

Matrix of a Linear Transformations

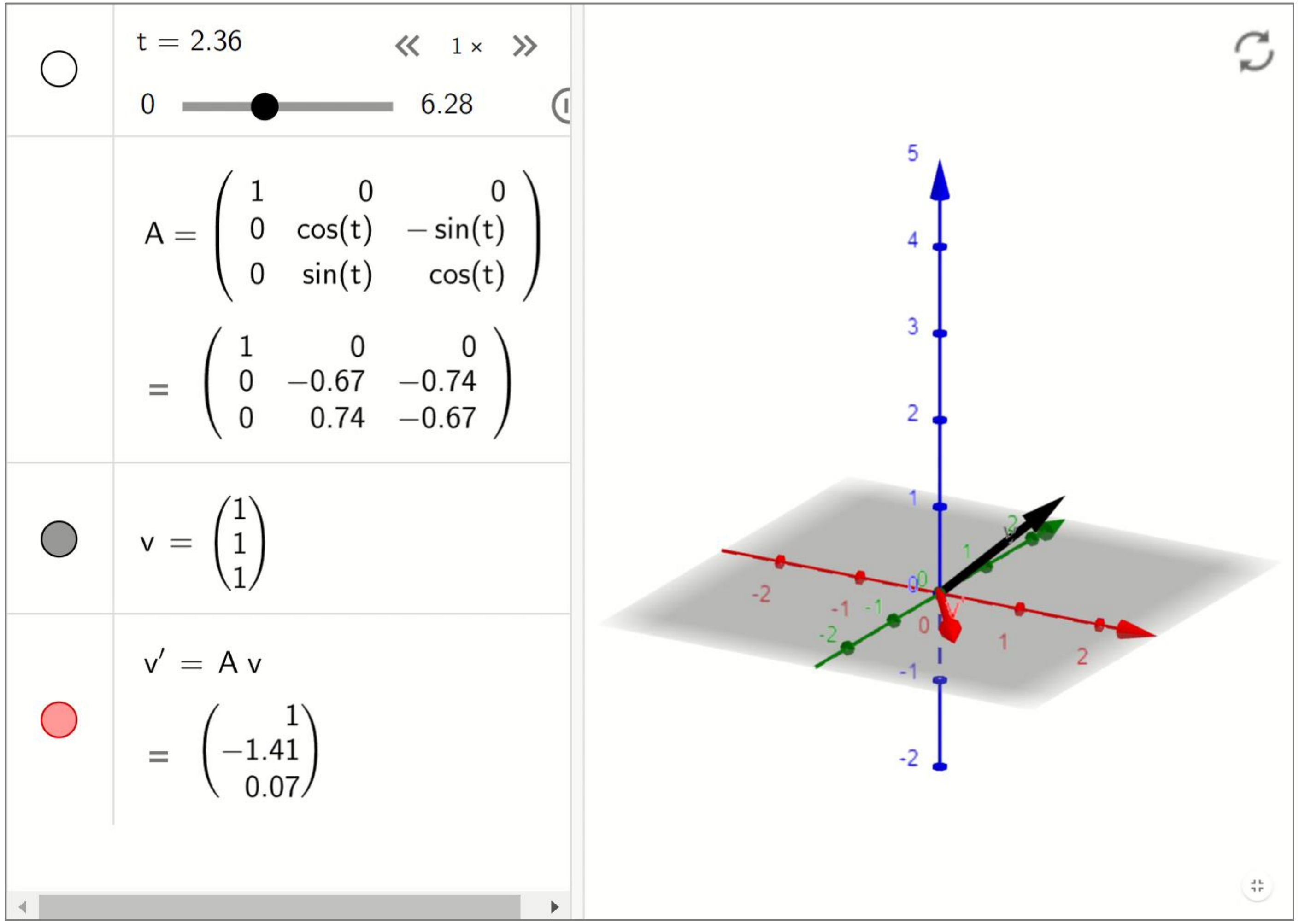
Rotation Transformation (Continued...)

3D rotation is not the same as 2D

For a 3D rotation,

we will have to specify:

- Angle of rotation
- Axis of rotation



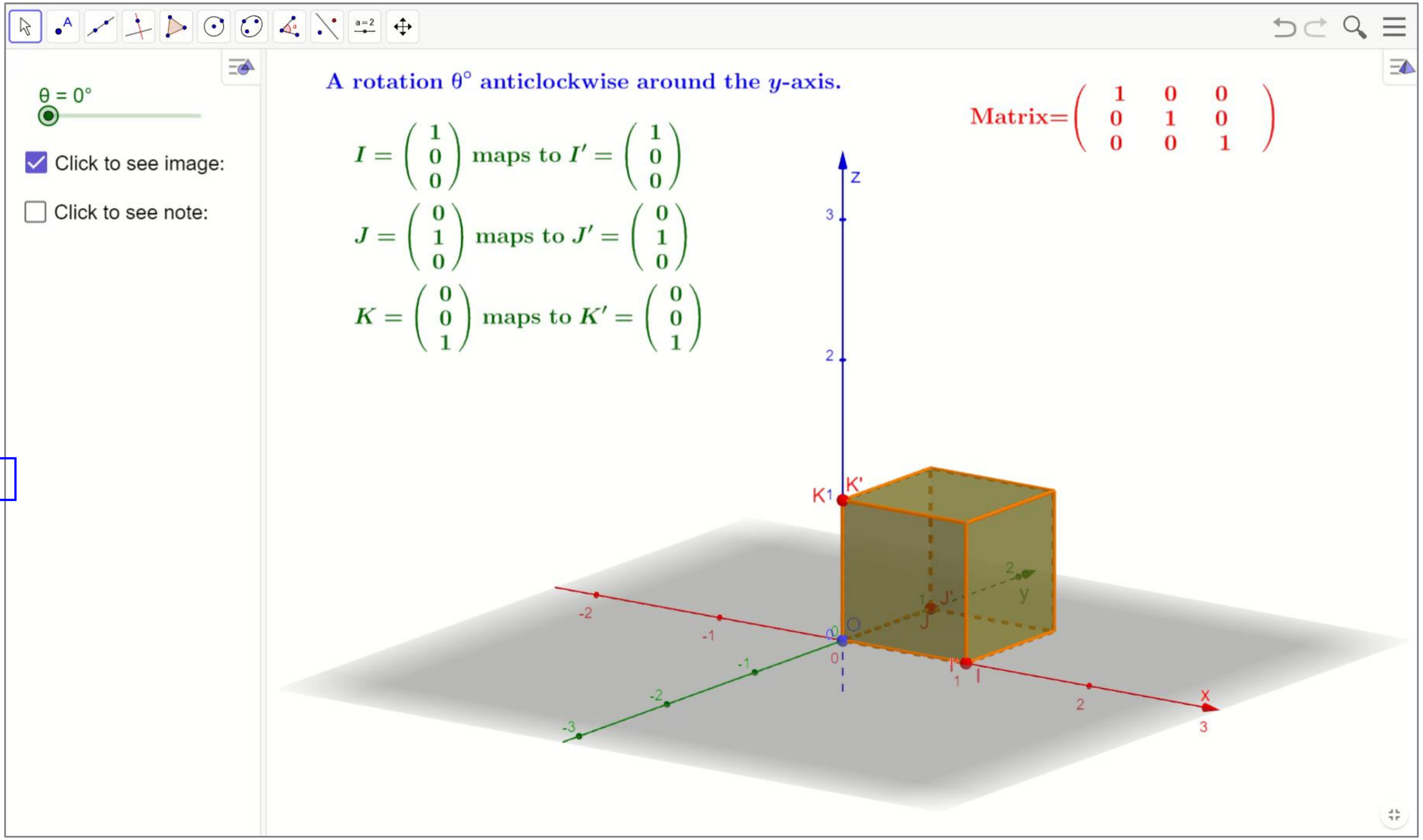
Matrix of a Linear Transformations

Rotation Transformation (Continued...)

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}$$

$$A = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}$$

$$A = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

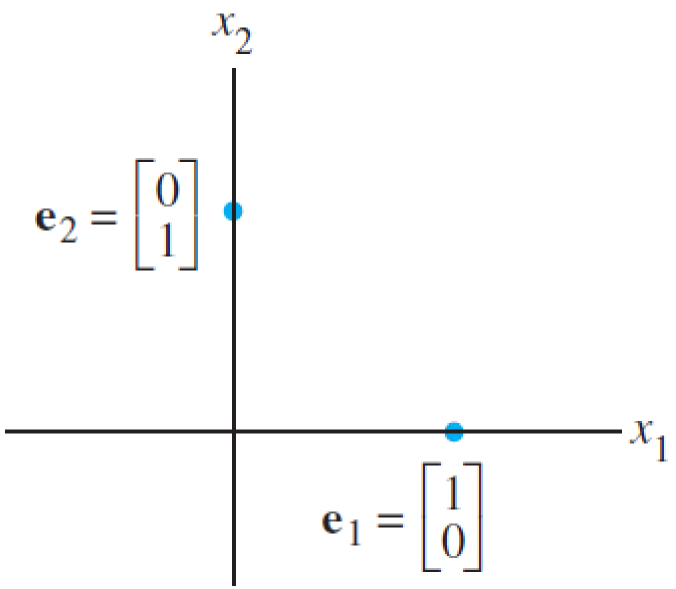


Standard Transformation Matrix

Finding Transformation Matrix A

- Every linear transformation from $\mathbb{R}^n \longrightarrow \mathbb{R}^m$ is actually a matrix transformation $\mathbf{x} \mapsto A\mathbf{x}$
- The key to finding A is to observe that T is completely determined by what it does to the columns of the $n \times n$ identity matrix I_n.
- Consider a 2×2 identity matrix I_2.

The columns of $I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ are $\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.



Standard Transformation Matrix

Finding Transformation Matrix A (Continued…)

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation. Then there exists a unique matrix A such that

$$T(\mathbf{x}) = A\mathbf{x} \quad \text{for all } \mathbf{x} \text{ in } \mathbb{R}^n$$

In fact, A is the $m \times n$ matrix whose jth column is the vector $T(\mathbf{e}_j)$, where $\mathbf{e}_j$ is the jth column of the identity matrix in $\mathbb{R}^n$:

$$A = \begin{bmatrix} T(\mathbf{e}_1) & \cdots & T(\mathbf{e}_n) \end{bmatrix}$$